NAMANGA SECONDARY SCHOOL

FORM FOUR — INFORMATION AND COMPUTER STUDIES

DATABASE PRACTICAL EXAMINATION

QUESTIONS PAPER — MICROSOFT ACCESS 2021

Time	3 Hours	Total Marks	100
Candidate Name	_____	Candidate No.	_____
Class / Stream	_____	Date	_____

GENERAL INSTRUCTIONS

1. Use **Microsoft Access 2021** to complete the practical.
2. Create and save the database as **NAMANGA_STUDENT_DATABASE.accdb**.
3. Create exactly three tables: **STUDENT, CLASS and ACTIVITY**.
4. Enter only the records provided in this paper.
5. Create only the relationship specified in Question C.
6. Use Design View, Lookup Wizard, Query Design, Forms and Reports where required.
7. Save every object using the exact name given in the question.
8. Test your work in Datasheet View and Print Preview before submission.
9. Answer all questions.

IMPORTANT DESIGN RULE: The database contains three tables, but only **CLASS.ClassCode** → **STUDENT.ClassCode** must be related. Do not create a relationship involving ACTIVITY.

QUESTION A — DATABASE CREATION

I. Open Microsoft Access 2021, create a blank database and save it as **NAMANGA_STUDENT_DATABASE.accdb**.

II. Create three tables named exactly **STUDENT**, **CLASS** and **ACTIVITY** using Table Design. Do not create any extra table.

QUESTION B — TABLE DESIGN AND DATA ENTRY

I. Create the **STUDENT** table with the following fields: StudentID, FirstName, LastName, Gender, DOB, ClassCode, Phone and Mark. Choose suitable Access data types, set StudentID as the primary key, and enter all ten student records below.

StudentID	FirstName	LastName	Gender	DOB	ClassCode	Phone	Mark
ST001	Amani	Mollel	Male	12/03/2010	F2A	0712345678	78
ST002	Neema	Joseph	Female	27/07/2011	F2A	0756789012	84
ST003	Musa	Kweka	Male	19/01/2009	F3A	0765432109	67
ST004	Halima	Said	Female	05/11/2010	F2B	0687654321	72
ST005	Baraka	Mushi	Male	22/09/2008	F3A	0711223344	91
ST006	Maria	John	Female	14/02/2012	F1A	0744556677	58
ST007	Kelvin	Mrema	Male	30/06/2011	F2B	0788990011	63
ST008	Rehema	Nnko	Female	17/08/2009	F3B	0622334455	88
ST009	Omari	Juma	Male	02/05/2010	F2A	0699887766	49
ST010	Asha	Mwakalinga	Female	25/12/2008	F3B	0714556677	76

II. Create the **CLASS** and **ACTIVITY** tables, set their primary keys, and enter all records exactly as shown.

ClassCode	ClassName	Stream	RoomNo	ClassTeacher
F1A	Form I	A	R01	Mr Kato
F2A	Form II	A	R02	Ms Neema
F2B	Form II	B	R03	Mr Baraka
F3A	Form III	A	R04	Ms Asha
F3B	Form III	B	R05	Mr Musa

ActivityID	ActivityName	Category	ActivityDate	Participants	Fee
AC01	Cyber Safety Talk	ICT	20/01/2026	80	0
AC02	Football Match	Sports	24/01/2026	96	500
AC03	Database Club	Academic	28/01/2026	24	0
AC04	Science Fair	Academic	13/02/2026	55	1000
AC05	Girls in ICT Day	ICT	28/02/2026	45	0
AC06	Netball Match	Sports	07/03/2026	60	300
AC07	ICT Workshop	ICT	21/03/2026	38	2000
AC08	Programming Challenge	ICT	28/03/2026	27	500

QUESTION C — RELATIONSHIP

I. In Database Tools → Relationships, create a relationship between **CLASS.ClassCode** and **STUDENT.ClassCode**. Make ClassCode in CLASS the primary key and ClassCode in STUDENT the foreign key. Set the relationship as one-to-many.

II. Enable **Enforce Referential Integrity**. Do not create any relationship involving ACTIVITY. Save the relationship layout.

QUESTION D — VALIDATION AND LOOKUP

I. In STUDENT, use Lookup Wizard for **Gender** with only **Male** and **Female**, and use Lookup Wizard for **ClassCode** to obtain values from the CLASS table.

II. Set these validation rules: **Mark** = Between 0 And 100; **Phone** = exactly 10 digits beginning with 0; **Participants** >= 1; **Fee** >= 0; **ActivityDate** must be within 2026.

QUESTION E — QUERIES

Query I — Q_AllStudents: Display StudentID, FirstName, LastName and Mark for all students.

Query II — Q_Performance: Display student name, Mark and a calculated PerformanceBand: 80 or above = Excellent; 60–79 = Good; below 60 = Needs Improvement.

Query III — Q_FormIII: Join STUDENT and CLASS and display FirstName, LastName, ClassName, Stream and Mark for Form III students only.

Query IV — Q_TotalCollection: Using ACTIVITY, calculate TotalCollection = Participants × Fee and display only records where TotalCollection > 50000.

Query V — Q_MinimumMark: Create a parameter query that asks the user to enter a minimum Mark and displays students whose Mark is greater than or equal to the entered value.

Query VI — Q_NameStartsM: Show students whose FirstName starts with M using Like.

Query VII — Q_LastNameAR: Show students whose LastName contains 'ar'.

Query VIII — Q_ClassIN: Show students in F2A, F2B or F3A using IN.

Query IX — Q_Mark75Plus: Show students with Mark >=75.

Query X — Q_Mark60_80: Show students with Mark between 60 and 80.

Query XI — Q_Female70Plus: Show female students with Mark >=70 using AND.

Query XII — Q_MaleOR90: Show students who are Male OR have Mark >=90.

Query XIII — Q_DOB_2009_2010: Show students born from 01/01/2009 to 31/12/2010.

Query XIV — Q_Older70Plus: Show students born before 01/01/2010 AND with Mark >=70.

Query XV — Q_Age16Plus: Create an Age field using DateDiff and display students aged 16 or above.

Query XVI — Q_ClassGender: Create a crosstab showing the number of students by ClassName and Gender.

Query XVII — Q_LargeLowCost: Show activities with Participants >=50 AND Fee <=1000.

Query XVIII — Q_ICT_Academic: Show activities from 01/02/2026 to 31/03/2026 where Category is ICT OR Academic.

Query XIX — Q_ActivityPattern: Show activities whose ActivityName starts with C OR ends with Day.

Query XX — Q_ParticipantsORFree: Show activities with Participants between 30 and 80 OR Fee = 0.

QUESTION F — FORMS

I. Create **FORM_STUDENT** from STUDENT using a **Columnar** layout. Create **FORM_CLASS** from CLASS using a **Tabular** layout.

II. Create **FORM_ACTIVITY** from ACTIVITY using a **Justified** layout. Open all forms and confirm that records can be viewed correctly.

QUESTION G — REPORTS

I. REPORT I — REPORT_STUDENT_CLASS: Use Q_FormIII as the record source. Show FirstName, LastName, ClassName, Stream and Mark. Group by ClassName and sort Mark from highest to lowest.

II. REPORT II — REPORT_ACTIVITY_CATEGORY: Use ACTIVITY as the record source. Group by Category and show ActivityName, Participants and Fee.

III. REPORT III — REPORT_PERFORMANCE: Use Q_Performance as the record source. Show student name, Mark and PerformanceBand. Add a clear report title and page number.

QUESTION H — TESTING AND FINAL CHECK

I. Test the database by attempting the following invalid values and record whether Access rejects them: Mark = 105; Participants = 0; Fee = -500; Gender = Other; Phone = 07123456; ClassCode = F9A. After testing, return the database to valid data.

II. Check the Navigation Pane and confirm that STUDENT, CLASS and ACTIVITY, the required saved queries, the three forms and the three reports are present. Preview all reports, make sure no information is cut off, save the database, and prepare **NAMANGA_STUDENT_DATABASE.accdb** for submission.

END OF QUESTION PAPER

Candidate should verify every object name before submission.